PHARMACEUTICAL QUALITY ASSURANCE

UNIT 1 NOTES

- QUALITY ASSURANCE
- TOTAL QUALITY MANAGEMENT
- ICH GUIDELINES
- QUALITY BY DESIGN
- ISO 9000 & 14000
- NABL ACCREDITATION

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Quality Management System

- QMS is a structured system of procedures and processes designed to ensure that an organization can consistently deliver products or services that meet customer & regulatory requirements.
- QMS is a set of guidelines & procedures in pharmaceutical industry for maintaining the quality & safety of their products & services.
- It is the act of supervising all the activities required for maintaining a desired level of quality.

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Concept Of Quality

- According to USFDA, "Quality is a measure of products or services ability to satisfy the customer's demand."
- Quality isn't only about the product being defect-free but also about fitness for use – i.e., how well it performs its intended function.
- In Pharma, quality means ensuring that the drug is safe, effective, pure & consistent throughout its shelf life.

Importance Of Quality

- Ensure safety & effectiveness of the product.
- Maintains uniformity & consistency in each batch.
- Builds customer trust & satisfaction.
- Helps in regulatory compliance (GMP, WHO guidelines etc.)
- Improves product reliability & shelf life.
- Enhances company reputation & market acceptance
- Supports cost controly by preventing rework & failures.

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QUALITY ASSURANCE

- Quality Assurance means making sure that a product is made correctly every time & is safe, effective, & of good quality.
- It focuses on preventing mistakes rather than finding them later.

OBJECTIVES OF QUALITY ASSURANCE

- To ensure good quality of the product
- To make products safe for use
- To maintain same quality in evey batch
- To follow rules & guidelines (GMP)
- To avoid errors & defects

SCOPE OF QUALITY ASSURANCE

Quality Assurance covers all steps from start to end:

- Raw Materials Checking
- Manufacturing process
- In- Process controls
- Testing of finished product
- Packaging, storage & distribution
- Proper documentation

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Elements Of Quality Assurance

① Good Manufacturing Practices

- GMP are basic rules that ensure products are made in a clean, safe & controlled environment.
- They include:
  - Clean buildings & equipment
  - Proper maintenance of machines
  - Qualified & trained workers
  - Correct manufacturing methods

② Documentation

- Documentation means writing down everything related to manufacturing & testing:
- It includes:
  - Standard Operating Procedures (SOPs)
  - Batch Manufacturing Records (BMR) etc.
- Proper documentation ensures traceability, helps in checking mistakes, & proves that work was done correctly.

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③ Quality Control

- Quality Control involves testing & checking raw materials, in-process materials & finished products.
- QC ensures that the product:
  - Meets required standards
  - Is safe & effective

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④ VALIDATION

- Validation is the process of providing that machines, methods & processes work properly.
- It confirms that:
  - Equipment gives correct results
  - Manufacturing process produces the same quality product every time.

⑤ TRAINING OF PERSONNEL

- All workers must be properly trained before doing their job.
- Training helps:
  - Workers follow SOPs correctly
  - Reduce human errors
  - Improves overall product quality

⑥ SELF - INSPECTION AND AUDITS

- Self inspection means checking the system regularly to find problems.
- Audits help in:
  - Finding mistakes early
  - Improving the quality system
  - Ensuring compliance with guidelines

IMPORTANCE OF QUALITY ASSURANCE

- Prevents poor- quality products
- Protects patient health
- Builds trust in the company
- Reduces wastage & looses

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Quality Control

- Quality Control is the part of quality management that checks & tests product to make sure they meet required standards.
- Its main role is to detect errors & defects in raw materials, in process materials, & finished products.

Objective Of Quality Control

- To ensure products meet quality specifications
- To detect defective or substandard materials
- To ensure safety & effectiveness of the product
- To maintain uniformity & consistency
- To support Quality Assurance activities

Scope Of Quality Control

Quality Control covers testing & checking activities such as:

- Testing of raw materials
- In-process testing during manufacturing
- Testing of finished products
- Stability testing
- Environmental monitoring

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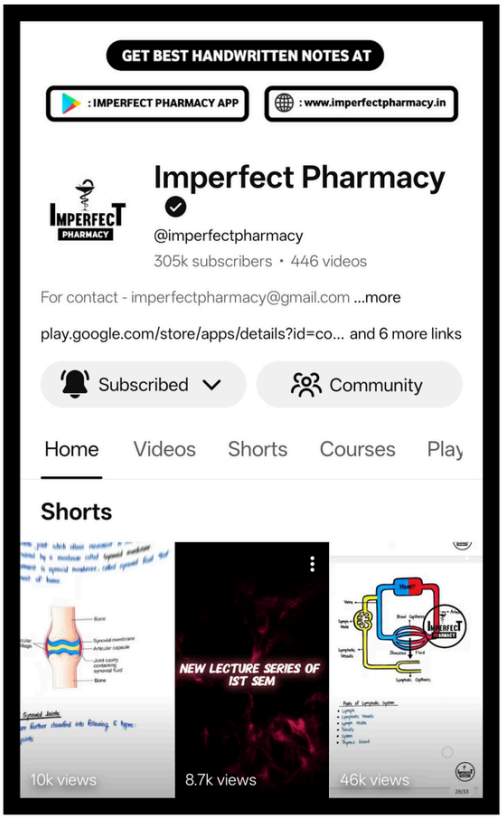

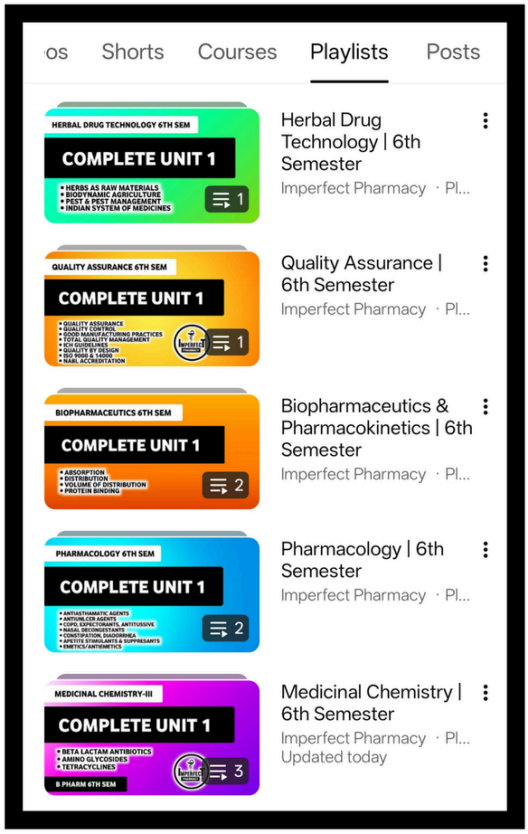

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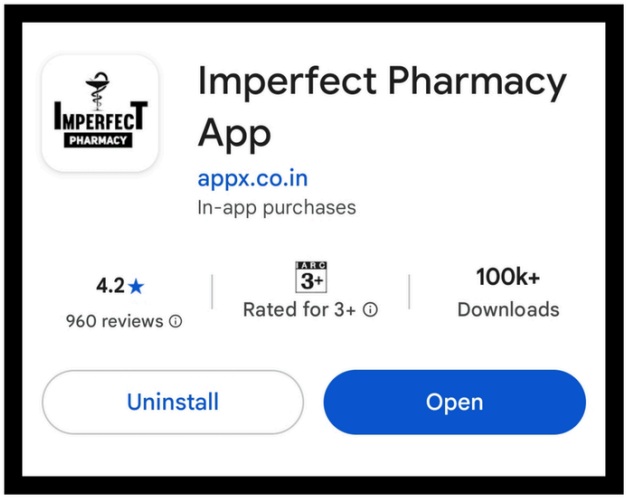

Elements Of Quality Control

① Raw Material Testing

- All raw materials are tested before use
- Ensures materials meet pharmacopoeial standards
- Prevents use of poor-quality materials

② In-Process Quality Control

- Testing done during manufacturing
- Checks weight variation, hardness, pH, moisture etc.
- Helps in early detection of errors

③ Finished Product Testing

- Final testing before product release
- Ensures product meets specifications for quality, safety & efficacy.
- Only approved products are released to the market.

④ Documentation

- Recording test results & observations
- Includes tests reports & analytical records
- Helps in traceability & audit purpose

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Important Of Quality Control

- Prevents release of defective products
- Ensures patient safety
- Maintains product consistency
- Helps in regulatory compliance
- Builds trust in the product

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Quality Assurance Vs Quality Control

Quality Assurance (QA)	Quality Control (QC)
• Focuses on process	• Focuses on product
• Prevents mistakes	• Finds mistakes
• Done before & during manufacturing	• Done after or during manufacturing
• Proactive approach	• Reactive approach
• Concerned with how work is done	• Concerned with what result is obtained
• Includes GMP, SOPs, validation, training	• Includes testing, analysis, inspection
• Responsibility of everyone	• Responsibility of QC department
• Aims to build quality	• Aims to check quality
• Reduces chances of defects	• Detects defects in products

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Good Manufacturing Process

- Good Manufacturing Practices (GMPs) are rules & guidelines that ensure medicines & other products are consistently made with good quality, safe & effective for use.
- GMP is a part of Quality Assurance.

Objectives Of GMP

- Ensure products are safe, effective, and of high quality.
- Maintain consistency in every batch
- Prevent contamination, mix-ups & errors
- Ensure compliance with regulatory guidelines
- Build trust among patients & consumers

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Elements Of GMP

① Premises & Equipment

- Factory should be clean, well-lit & well-ventilated.
- Equipment should be properly maintained & calibrated.
- Layout should prevent contamination & mix-ups

② Personnel And Training

- Staff should be qualified & trained
- Follow hygiene & safety rules
- Proper supervision & clear responsibilities

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③ Raw Materials And Packaging

- Materials should be tested before use.
- Stored in proper conditions
- Prevents use of poor-quality materials.

④ Production Process

- Follow standard operating procedures (SOPs)
- Maintain in-process checks
- Ensure uniformity & consistency

⑤ Quality Control

- Testing of raw materials, in-process samples & finished products.
- Only products meeting specifications are released for sale.

⑥ Documentation

- Maintain records of every step
- Includes batch records, SOPs, equipment logs
- Helps in traceability, audit & investigation

⑦ Storage And Distribution

- Products stored under correct conditions (temperature, humidity)
- Proper labelling & handling during distribution.

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Importance Of GMP

- Ensures patient safety
- Maintains product quality & consistency
- Reduces wastage, recalls & legal issues
- Helps in regulatory compliance
- Builds reputation & trust

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TOTAL QUALITY MANAGEMENT

- Total - Made up of the whole (or) Complete.
- Quality - Degree of Excellence a product or service provides to the customer in present & future.
- Management - Act, art, or manner of handing, controlling, directing, etc. TQM is the art of manging the whole to achieve excellence.

- TQM requires that the company maintain this quality standard in all aspects of its business. This requires ensuring that things are done right the first time & that defects and waste are eliminated from operations.

OBJECTIVES OF TQM

- To achieve customer satisfaction.
- To ensure continuous improvement in all activities.
- To involve all employees in quality improvement.
- To reduce wastage & defects in processes.
- To enhance efficiency & productivity.
- To maintain long-term success through cosistent quality.

PRINCIPLE OF TQM

- Total Quality Management is built on a set of core principles that help organizations achieve continuous improvement & customer satisfaction.

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① Focus On Customer

- The main goal of TQM is to meet or exceed customer expectations.
- Customer satisfaction determines product quality & business sucess.
- Regular feeback helps identify areas for improvement.

② Employee Involvement

- Every employee is responsible for maintaining quality.
- Encourages teamwork, motivation, and participation in decision-making.
- Proper training & empowerment help employees contribute to quality goals.

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③ Process Centered

- Quality is achieved by improving the process, not just checking the final product.
- Each step in production is monitored & optimized to prevent errors.
- A controlled process ensures consistent & reliable output.

④ Integrated System

- All department (production, QC, QA, HR, marketing) work together as one system.
- Quality is everyone's resposibility – not limited to a single department.
- Promotes coordination & communication across the organization.

⑤ Strategic And Systematic Approach

- Quality improvement must be planned strategically.
- A clear vision, mission, & quality policy guide the organization.
- Quality objectives are set, implemented, & reviewed systematically.

⑥ Decision-Making Based On Facts

- Decisions should rely on accurate data, not assumptions.
- Statistical tools & quality records help analyze performance & identify issues.
- Data-driven actions improve reliability & reduce defects.

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⑦ Communication

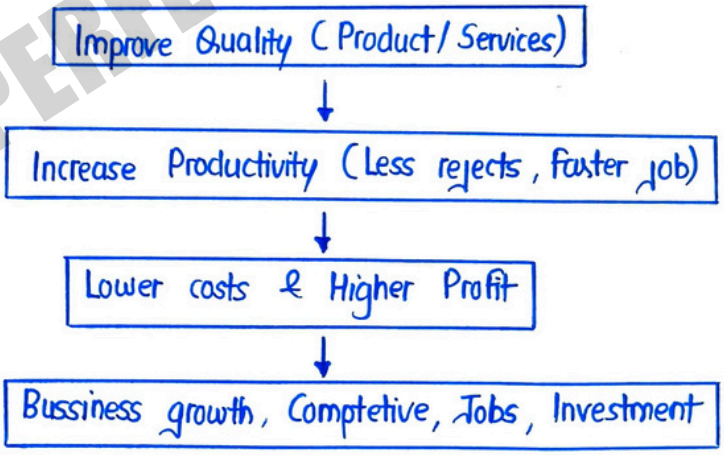
- Open & effective communication ensures that quality goals are understood by all.
- Promotes trust, cooperation & problem-solving among employees.
- Regulars meetings & feeback strengthen teamwork.

⑧ Continuous Improvement

- T&M emphasizes ongoing efforts to improve processes, products & services.
- The aim is to reduce errors, improve efficiently & enhance customer satisfaction.

Effect

- The flowchart below shows the effect of T&M:



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Benefits Of TQM

- Improves product quality & reliability.
- Increases customer satisfaction & loyalty.
- Reduces waste, defects & production cost.
- Enhances efficiency & productivity.
- Promotes employee involvement & teamwork.
- Ensures regulatory compliance.
- Improves brand image & market competetiveness
- Leads to long-term success & profitability.

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ICH

- ICH stands for "International Council of Harmonisation" of technical requirements for pharmaceuticals for human use.
- It is an international organization that brings together regulatory authorities & pharmaceutical industry to develop common guidelines for any drug development & registration.
- ICH makes common rules for medicines so that same quality, safety & effectiveness standards are followed in all countries.

History & Need Of ICH

- ICH was formed in 1990 by representatives from the regulatory agencies & industries of US, Europe & Japan.
- Initially, it was known as "International Conference of Harmonisation" & Now, it's known as "International Council of Harmonisation".
- The ICH Secretriat (Administrative office) is located in Geneva, Switzerland.
- Before the ICH existed (pre-1990), different countries had different rules for testing drugs.
- If a company wanted to sell a medicine in other countries such as, US, Europe & Japan, they often had to repeat the same expensive tests two-three times just to meet different paperwork requirements.
- The ICH was formed to "hormonize" these rules so that one set of high-quality standards would be accepted everywhere.

Objectives Of ICH

- Harmonize (make same) drug regulations worldwide.
- Ensure quality, safety & efficacy of medicines.
- Reduce unneccessary clinical trials.
- Faster availability of medicines to patients.

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MEMBERS IN ICH

- The founding members represents authorities & research based industries in European Union, Japan & USA.

① JAPAN: MHLW (Ministry of Health, Labour & Welfare)
JPMA (Japan Pharmaceutical Manufacturers Association)

② EUROPE: EC (European Commission)
EFPIA (European Federation of Pharmaceutical Industries & Association)

③ USA: FDA (Food & Drug Administration)
PhRMA (Pharmaceutical Reseach & Manufactures of America)

- Additional members include observes such as WHO & regulatory members from countries other than Japan, Europe & USA.

ICH GUIDELINES

ICH Guidelines are divided into 4 main series:

- Q : Quality
- S : Safety
- E : Efficacy
- M : Multidiscipilinary

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QUALITY GUIDELINES

- The ICH Q-series guidelines deals with the 'Quality' of pharmaceutical products.
- It ensures that a medicine is consistently manufactured, stable, pure & safe for patient use throughout its shelf life.
- These guidelines cover:
  - Raw Materials
  - Manufacturing Process
  - Testing Methods
  - Stability
  - Quality Systems

OBJECTIVES OF ICH Q-Series

- To ensure high quality of drugs
- To maintain batch-to-batch consistency
- To control impurities & degradation products
- To establish international quality standards
- To protect patient safety

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Quality Guidelines	
Q1A - Q1F	Stability
Q2	Analytical Validation
Q3A - Q3D	Impurities
Q4A - Q4B	Pharmacopoeias
Q5A - Q5E	Quality of Biotechnological Product
Q6A - Q6B	Specifications
Q7	Good Manufacturing Process
Q8	Pharmacopoeial Development
Q9	Quality Risk Management
Q10	Pharmaceutical Quality System
Q11	Development & Manufacture of Drug Substances
Q12	Lifecycle Management
Q13	Continuous Manufacturing of Drug Substances & Drug Products
Q14	Analytical Procedure Development

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Q1 - Stability Testing

- Purpose : To study how long a drug remains stable & effective under different conditions

- Includes :
  - Q1A - Stability testing of new drug substances & products
  - Q1B - Photostability (effect of light)
  - Q1C - Stability testing of new dosage forms
  - Q1D - Bracketing & Matrixing designs
  - Q1E - Evaluation of stability data

- Helps in diciding expiry date & storage conditions.

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Q2 - Analytical Validation

- Purpose: To ensure analytical test methods are accurate, reliable & reproducible.

- Checks: - Accuracy
  - Precision
  - Specificity
  - Linearity

- Ensures correct testing of drug identity, purity & strength.

Q3 - Impurities

- Purpose: To control harmful impurities in drugs.

- Types: - Q3A: Impurities in drug substances
  - Q3B: Impurities in drug products
  - Q3C: Residual solvents
  - Q3D: Elemental impurities
  - Q3E: Extractables & leachables

- Ensures drugs are safe & free from toxic impurities.

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Q4 - Pharmacopoeial Harmonisation

- Purpose: To harmonize standards of different pharmacopoeia like USP, BP, JP, IP.
- Same quality standards accepted in different countries.

Q5 - Quality Of Biotechnological Products

- Purpose: To ensure quality of biological medicines like insulin & vaccines.
- Covers: - Viral safety
  - Stability
  - Cell Substances
  - Manufacturing process
- Important for biotech & biological products.

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Q6 - Specifications

- Purpose: To define test procedures & acceptance criteria for drugs.
- Decides what tests to perform & what limits are acceptable.

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Q7 - Good Manufacturing Practice (GMP)

- Purpose: To provide GMP guidelines for Active Pharmaceutical Ingredients (APIs).
- Covers:
  - Personnel
  - Equipment
  - Documentation
  - Hygiene
  - Quality Control
- Ensures safe & consistent manufacturing.

Q8 - Pharmaceutical Development (PD)

- Purpose: To develop medicines using Quality by design (QbD) approach.
- Quality is built during development, not checked only at the end.

Q9 - Quality Risk Management

- Purpose: To identify, evaluate, & control risk related to drug quality.
- Helps prevent quality failures & recalls.

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Q10 - Pharmaceutical Quality System (PQS)

- Purpose: To maintain quality throughout the entire product life cycle.
- Includes manufacturing, change control & continuous improvement.

Q11 - Development & Manufacture Of Drug Substance

- Purpose: Guidelines for development & manufacture of chemical & biological drug development.

Q12 - Life Cycle Management

- Purpose: To manage post-approval changes in pharmaceutical products.
- Ensures continuous quality after approval.

Q13 - Continuous Manufacturing

- Purpose: Guidelines for continuous manufacturing of drug substances & products
- Improves efficiency & quality consistency.

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Q14 - Analytical Procedure Development

- Purpose: Guidelines for developing & improving analytical procedures.
- Supports better analytical validation.

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Safety Guidelines

- Safety guidelines ensure that a drug is safe for human use.
- It includes study of toxicity of drugs in animals.
- Check effects on organs, DNA, Cancer risk & reproduction.
- Ensure the drug doesn't cause serious harmful effects.

Efficacy Guidelines

- Efficacy guidelines ensure that a drug works effectively in patients.
- Guide clinical trials in humans
- Include Good Clinical Practice (GCP)
- Study proper dose, effectiveness, & side effects

Multidisciplinary Guidelines

- Multidisciplinary guidelines cover common regulatory topics.
- Provides CTD format for drug submission
- Standardize electronic data & medical terms
- Help smooth communication b/w regulatory authorities & researches

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STABILITY TESTING

- Stability means the ability of a drug substance or drug product to remain chemically, physically & therapeutically stable throughout its shelf life when stored under recommened conditions.
- Stability testing checks how long a drug remains good & effective.
- It studies effect of temperature, humidity & light.
- It helps to decide expiry date & storage conditions.

TYPES OF STABILITY TESTING

① LONG-TERM STABILITY STUDY

- Done under normal storage conditions
- Helps to decide shelf life.

② ACCELERATED STABILITY STUDY

- Done under high temperature & humidity
- Predicts long-term stability in short time

③ INTERMEDIATE STABILITY STUDY

- It performed when drug fails under accelerated conditions

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Storage Conditions

Study Type	Storage Condition
Long - term	25°C/60 or 30°C/65% RH
Intermediate	30°C/65% RH
Accelerated	40°C/75% RH

Climate Zones

- ICH devides the world into climate zones based on temperature & humidity:

  - Zone I - Temperate
  - Zone II - Mediterranean
  - Zone III - Hot & Dry
  - Zone IV - Hot & Humid

- India belongs to Zone IV

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ICH Guidelines For Stability Testing

- ICH provides guidelines for stability testing of pharmaceutical products under Q1 of Q series as follows :

① Q1A - Stability Testing of New Drug Substances & Products

- Used for new drugs
- Drug is stored under different conditions
- Long-term, intermediate & accelerated studies are done
- Helps to fix shelf life & expiry date

② Q1B - Photostability Testing

- Checks effect of light on drug
- Finds whether drug is light sensitive.
- Helps to detect light-protecting packaging.

③ Q1C - Stability Testing for New Dosage Forms

- Used when a new dosage form is made
- Example: Tablet changed to syrup.
- Ensures new dosage form remains stable

④ Q1D - Bracketing & Matrixing

- Used to reduce number of stability tests
- Bracketing tests extreme samples only
- Matrixing tests selected examples
- Saves time & cost

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Q1E - Evaluation of Stability Data

- Stability results are analysed
- Helps to calculate expiry dates
- Uses statistical methods

Q1F - Stability Data for Climate Zone III & IV

- Used for Hot & humid countries
- India comes under Zone IV
- Ensures drugs remain stable in Indian climate

IMPORTANCE OF ICH STABILITY GUIDELINES

- Ensures drugs remain safe & effective till expiry.
- Helps regulatory approval
- Prevents product degradation during storage & transport
- Ensures uniforms standards worldwide.

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Quality By Design

- The concept of QbD was mentioned in the ICH Q8 guideline, which states that "quality can't be tested into products, i.e., quality should be built in by design".
- Quality by design (QbD) is a systematic, scientific, and risk-based approach to pharmaceutical development that ensures predefined product quality by understanding the process & controlling the variables.

Concept

- QbD focuses on building quality into the product from the beginning, rather than testing it at the end.
- It emphasizes "Quality can't be tested into products, it should be built in.
- It links product performance with process parameters through scientific understanding.

Objectives Of QbD

- To ensure consistent product quality & efficacy.
- To reduce variability in manufacturing.
- To minimize risk of product failure.
- To enhance process understanding & control.
- To facilitate regulatory flexibility & faster approvals.

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Elements Of QbD

The following Elements can be included in the study of QbD:

- QTPP
- CQAPs
- Risk Assesment
- Design Space
- Control Strategy
- PAT
- Continuous Improvement

① Quality Target Product Profile (QTPP):

- Describes the intented use, safety, efficacy & quality of the final product.
- Examples: Table strength, dissolution rate, stability, etc.

② Critical Quality Attributes (CQAs):

- Physical, chemical, biological, or microbiological properties that must be within limits for product quality.
- Example: Assay, Content uniformity, hardness, disintegration time.

③ Risk Assesment (RA):

- Identifies & evaluates risks to product quality.
- Tools: FMEA (Failure Mode and Effect Analysis), Ishikawa (Fishbone) diagram, Risk Matrix.

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④ Design Space

- The multidimentional combination of input variables (like formulation and process parameters) that ensures product quality.

⑤ Control Strategy

- A set of controls & monitoring parameters to maintain process within design space.
- Includes raw material testing, in-process controls, & end-product testing.

⑥ Process Analytical Technology (PAT):

- A tool used inproduct quality.
- Working within this space is not considered a change by regulatory agencies.
- QbD for real-time monitoring of manufacturing process.
  Ensures consistent product quality without stopping the process.

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⑦ Continuous Improvement

- Data from manufacturing is continuosly analyzed for process optimization & innovation.

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Steps In QbD Approach

- Define QTPP
- Identify CQAs
- Perform risk assesment
- Design & conduct experiments.
- Establish design space
- Develop control strategy
- Validate & implement continuous improvement.

Benefits Of QbD

- Ensure consistent product quality.
- Improves process understanding.
- Reduces batch failures & variations.
- Provides regulatory flexibility.
- Saves cost & development time.
- Enables effective risk management.
- Support continuous improvement.
- Increases patient safety & reliability.

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ISO 9000

- ISO 9000 is a series of international standards developed by the International Organization for Standardization (ISO) that provide guidelines & requirements for establishing & maintaining an effective Quality Management System (QMS).
- In industrial pharmacy, ISO 9000 principles are applied to ensure that pharmaceutical products are consistently manufactured and controlled meet quality, safety & efficacy requirements.

Objectives Of ISO 9000

- Ensure consistent product quality
- Improve customer satisfaction
- Establish effective quality management system
- Promote continuous improvement
- Ensure compliance with regulatory requirements
- Encourage evidence-based decision making.

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ISO 9000 Family Of Standards

The ISO 9000 series include several related standards:

Standards	Title / Focus
ISO 9000:2015	Fundamentals & vocubalary of QMS
ISO 9000:2015	Requirement for quality management system (certified stadard)
ISO 9004:2018	Guidelines for achieving sustained access through QMS
ISO 19011:2018	Guidelines for auditing management systems

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History Of ISO 9000

- Originally published in 1987 by the International Organization for Standardization (ISO).
- Major revisions occured in :
  - 1994 : Focused on preventive actions.
  - 2000 : Combined 9001, 9002, and, 9003 into a sigle standard (ISO 9001).
  - 2008 : Improved clarity & compatibility with ISO 14001.
  - 2015 : Introduced risk-based thinking & leadership focus.
- The ISO 9000 family now includes ISO 9000, 9001, 9004 & 19011 standards.

ISO 9000 Principles

- The ISO 9000 : 2015 & ISO 9001: 2015 standard are based on seven quality management principles that senior management can apply for organizational improvement.

① Customer focus
② Leadership
③ Engagement of people
④ Process approach
⑤ Improvement
⑥ Evidence-based decision making
⑦ Relationship management

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① Customer Focus

- The main goal is to meet & exceed customer expectations.
- Satisfied customers ensure long-term success & trust in organization.

② Leadership

- Strong leadership establishes a clear vision & direction.
- It helps align people & processes toward achieving quality objectives.

③ Engagement Of People

- Involving all employees increases motivation & responsibility.
- Everyone contributes to process improvement & product quality.

④ Process Approach

- Managing activities as interrelated processes improves efficiency & consistency.
- It ensures predictable & desired outcomes.

⑤ Improvement

- Continuous improvement is essential for maintaining performance & competetiveness.
- It encourages innovation & adaptation to changes.

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⑥ Evidence - Based Decision Making

- Decision should be based on data and analysis, not assumptions.
- This reduces errors & improves effectiveness of actions.

⑦ Relation Management

- Good relationship with suppliers, customers & stakeholders enhance cooperation.
- It supports sustained success & value creation.

Benefits Of Implementing ISO 9000 In Industrial Pharmacy

- Ensures consistent product quality & compliance with GMP.
- Promotes process standardization across departments & facilities.
- Enhances documentation & traceability, aiding regulatory inspections.
- Encourages a quality - oriented organizational culture.
- Provides a framework for continual improvement.
- Faciliates international market access due to globally recognized certification.

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ISO 14000

- The ISO 14000 series is a set of international standards developed by International Organization for Standardization (ISO) that focus on Environmental Management System (EMS).
- These standards help organizations minimize negative environmental impacts from their operations, comply with environmental laws & regulations, and persue continuous environmental performance improvement.
- In industrial pharmacy, ISO 14000 provides a structured approach for managing environmental responsibilities — such as waste disposal, energy consumption, chemical handling & pollution control — while maintaining high standards of quality & safety.

Objectives Of 14000 ISO In The Pharmaceutical Industry

- To control & reduce environmental pollution arising from pharmaceutical operations.
- To ensure compliance with national & international environmental regulations.
- To promote efficient use of natural resources & minimize waste.
- To prevent contamination from hazardous chemicals & effluents.
- To demonstrate corporate environmental responsibility & transparency.
- To integrate environmental management into the overall EMS.

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ISO 14000 Environmental Standards

ISO Standard	Standard Name/Description
ISO 14001:2015/Amend 1:2024	Environmental Management Systems – Requirements with guidance for use
ISO 14004:2016	General guidelines on implementation
ISO 14005:2019	Guidelines for a flexible approach to phased implementation
ISO 14006:2020	Guidelines for incorporating ecodesign
ISO 14015:2022	Environmental assessments of sites and organizations
ISO 14020:2020	Environmental labels and declarations – General principles
ISO 14024:2018	Type I environmental labelling – Principles and procedures
ISO 14025:2006	Type III environmental declarations – Principles and procedures
ISO 14031:2013	Environmental performance evaluation – Guidelines
ISO 14040:2006	Life cycle assessment – Principles and framework
ISO 14044:2006	Life cycle assessment – Requirements and guidelines
ISO 14046:2014	Water footprint – Principles, requirements and guidelines
ISO 14050:2020	Environmental management – Vocabulary
ISO 14063:2020	Environmental communication – Guidelines and examples
ISO 14064-1:2018	Greenhouse gases – Org. quantification and reporting
ISO 14064-2:2019	Greenhouse gases – Project-level GHG reductions
ISO 14064-3:2019	Greenhouse gases – Verification/validation guidance
ISO 14067:2018	Greenhouse gases – Product carbon footprint

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Key Principle Of ISO 14000

- The ISO 14000 series operates on several guiding principles:

① Environmental Policy Commitment - Defining the organization's intent to protect the environment.

② Prevention over Correction - Focusing on preventing pollution rather than reacting to it.

③ Compliance with regulations - Adhering to environment laws & standards.

④ Continuous improvement - Setting & achieving progressively higher environmental goals.

⑤ Stakeholder Involvement - Engaging employees, supplies, customers & communities.

⑥ Documentation & Transparency - Maintaing traceable & auditable environmental records.

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Objectives

- Improves environmental performancy by reducing pollution & waste.
- Helps Organization follow environmental laws & regulations.
- Reduces operational costs by saving energy & resources.
- Enhances company image & reputation.
- Promotes sustainable & eco-friendly practices.
- Encourages continuous improvement in environmental management.
- Supports international trade through global recognization.

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NABL

- The National Accreditation Board for Testing and Callibration Laboratories (NABL) is an autonomus body under the Quality Council of India (QCI) that provides accreditation to laboratories performing testing, calibration & medical diagnostics in India.
- NABL play a vital role in ensuring that laboratories involved in pharmaceutical industry, calibration of equipment & quality control operate in accordance with international standards for competence & reliability.
- Accredation by NABL signifies that laboratory is technically competent, follows validated methods, and produces accurate, reliable & traceable results.

Historical Background

- NABL was established in 1988 under the Department of Science and Technology (DST), Government of India.
  Later, it became a constituent board of the Quality Council of India.
- NABL is a signatory to international mutual recognition arrangements:
  - Asia Pacific Accredetation Cooperation (APAC) MRA
  - International Laboratory Accredetation Cooperation (ILAC) MRA
  - These recognitions make NABL - accredited laboratory results globally accepted.

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OBJECTIVE OF NABL

- To provide a formal recognition system for laboratories that meet international standards of competence.
- To promote quality assurance & technical competence in testing & calibration.
- To ensure data integrity & traceability in laboratory results.
- To enhance acceptance of Indian Laboratory data, globally, minimizing the need for retesting.
- To facilitate regulatory compliance in industries such as pharmaceuticals, food, environment, & healthcare.

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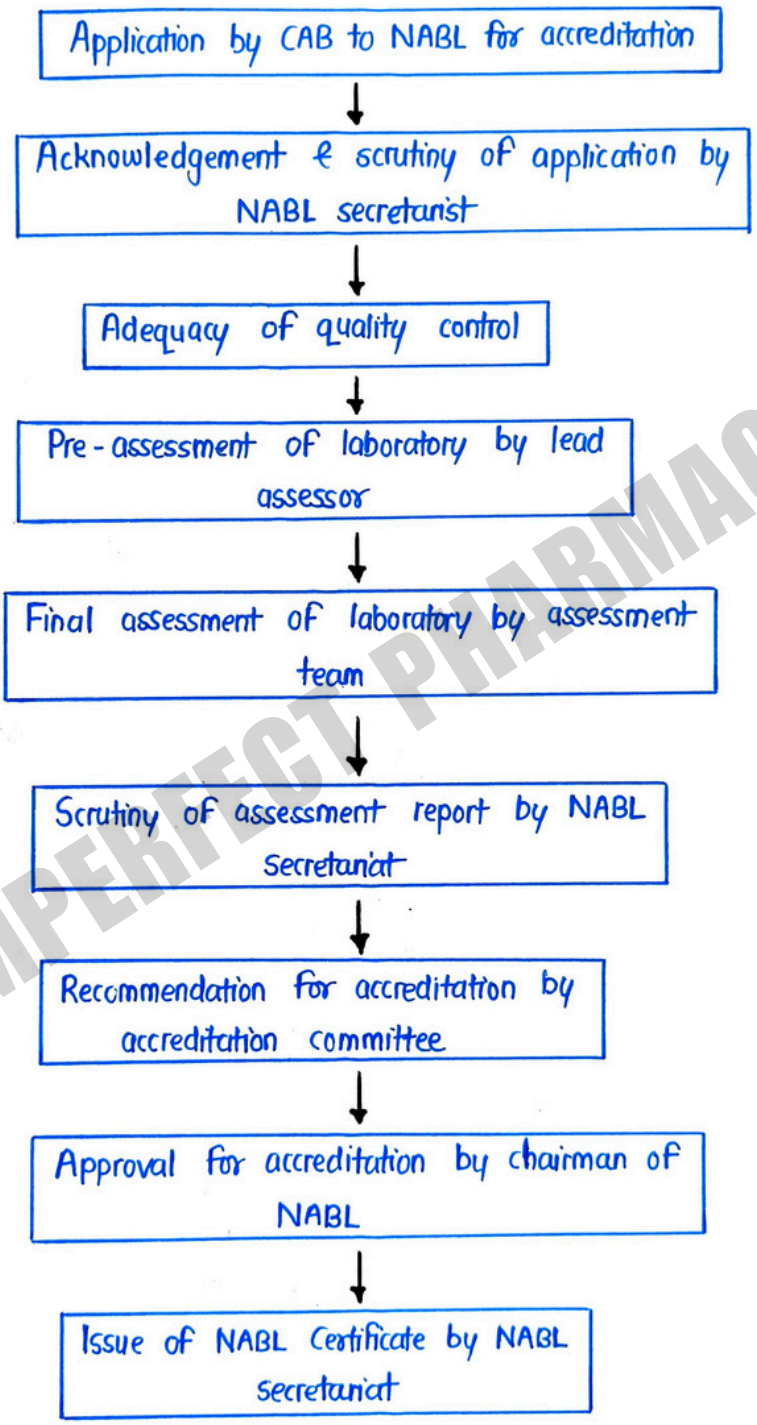
NABL Accreditation Process



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Benefits Of NABL Accreditation

- Ensures laboratory follows international quality standards.
- Increases reliability & accuracy of test results.
- Builds customer confidence in lab reports.
- Reduces need for re-testing of samples.
- Helps in national & international recognition.
- Improves overall laboratory performance & competence.
- Encourages continuous improvement through regular audits.

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THANK YOU

FOR CHOOSING IMPERFECT PHARMACY AS YOUR STUDY PARTNER

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